

2022/2023 TUTORIALS & QUESTIONS BANK

Class: HND II

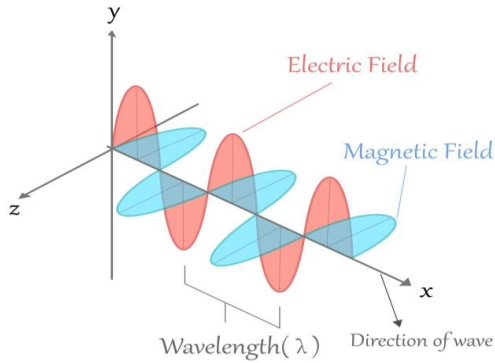
Module Code: EEE 445

Module Title: DIGITAL COMMUNICATION III

INSTRUCTION: Section A: OBJECTIVES & Section B: Essay
It worth trying to answer all questions in Section A, and B

SECTION A: OBJECTIVES

Use the EM wave signal information below to answer Q.1- 4



Q1. Wave direction of propagation is...

- A. All listed plane below
- B. z direction
- C. y direction
- D. x direction

Q2. Which of these plane lies the Electric Field Component (E)

- A. All listed plane above
- B. z plane
- C. x plane
- D. y plane

Q3. Which of the plane lies the Magnetic Field Component (H)

- A. All listed plane above
- B. y plane
- C. x plane
- D. z plane

Q4. Which of the following is NOT true about the properties of an electromagnetic wave?

- A. Have both electric (E) & magnetic (M) components.
- B. Affected by the medium through which its travels
- C. Travel in free-space or guided path (waveguide)
- D. Increases as the distance increases

Q5. Which of the followings laws is NOT applies to EM wave...

- A. Biot Savat's law
- B. Maxwell's law
- C. Faradays law
- D. Newton's law

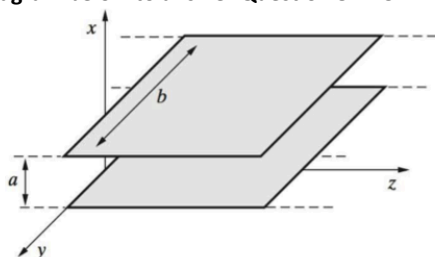
Q6. The expression $E_y(x, y, z, t) = E_y(z, t)$ shows the wave has no dependence in...

- A. x or t
- B. y or t
- C. x or z
- D. x or y

Q7. Which of the followings is parameters of EM wave...

- A. Wavelength λ
- B. Frequency f
- C. Radial frequency ω
- D. All the above

Use the diagram below to answer Question 8 – 15:



Q8. For an ideal parallel-plate waveguide, the plates are assumed to be infinite in extent in...

- A. x and z directions.
- B. x and y directions.
- C. y and z directions.
- D. All listed plane above

Q9. In an ideal parallel-plate waveguide, material between the plates is assumed to...

- A. be free.
- B. have losses
- C. be lossless.
- D. All listed plane above

Q10. The conductor plates are finite along which direction?

- A. z direction
- B. y direction
- C. x direction
- D. All listed plane above

Q11. The plates are infinite along y direction with condition...

- A. $b = a$
- B. $a \gg b$
- C. $b \gg a$.
- D. All the above

Q11. Propagation is governed by which of the law/equations?

- A. Biot Savat's law
- B. Faradays law
- C. Maxwell's law
- D. Gauss's law

Q12. For perfectly conducting plates with source-free, simple, and lossless assumption. Which of the following applies?

- A. ϵ and μ non-constants and $\alpha = 0$, μ and ϵ complex
- B. ϵ and μ are complex and $\alpha = 0$, μ and ϵ are real
- C. ϵ and μ are constants and $\alpha = 0$, μ and ϵ are real
- D. None of the above

Q13. In waveguides, the equations are subject to which of the following boundary conditions at the conductor surfaces?

- A. $E_{Tangential} \neq 0$ $H_{Normal} = 0$
- B. $E_{Tangential} = 0$ $H_{Normal} \neq 0$
- C. $E_{Tangential} = 0$ $H_{Normal} = 0$
- D. $E_{Tangential} \neq 0$ $H_{Normal} \neq 0$

Q14. For wave equation $\bar{\gamma} = \bar{\alpha} + j\bar{\beta}$, with $\bar{\alpha}$ and $\bar{\beta}$ being real, $\bar{\gamma} = \bar{\alpha}$ (i.e., when $\bar{\beta} = 0$) denote which of the following?

- A. Electromagnetic wave
- B. Propagating wave
- C. Evanescent wave
- D. None of the above

Q15. For wave equation $\bar{\gamma} = \bar{\alpha} + j\bar{\beta}$, with $\bar{\alpha}$ and $\bar{\beta}$ being real, $\bar{\gamma} = j\bar{\beta}$ (i.e., when $\bar{\alpha} = 0$) denote which of the following?

- A. Electromagnetic wave
- B. Propagating wave
- C. Evanescent wave
- D. None of the above

Answer Q.16 – 12 using field solutions for TE, TM & TEM mode.

Q16 Transverse Electric Mode: which of the following is true?

- A. $E_z \neq 0$, $H_z = 0$
- B. $E_z \neq 0$, $H_z \neq 0$
- C. $E_z = 0$, $H_z \neq 0$
- D. $E_z = 0$, $H_z = 0$

Q17. Transverse Magnetic Mode: which of the following is true?

- A. $E_z = 0$, $H_z = 0$
- B. $E_z = 0$, $H_z \neq 0$
- C. $E_z \neq 0$, $H_z = 0$
- D. $E_z \neq 0$, $H_z \neq 0$

Q18. Transverse Electromagnetic (TEM) Mode: which of the following is true?

- A. $E_z \neq 0$, $H_z = 0$
- B. $E_z = 0$, $H_z \neq 0$
- C. $E_z = 0$, $H_z = 0$
- D. $E_z \neq 0$, $H_z \neq 0$

Q19. Hybrid Mode: which of the following is true?

- A. $E_z \neq 0$, $H_z = 0$
- B. $E_z = 0$, $H_z \neq 0$
- C. $E_z \neq 0$, $H_z \neq 0$
- D. $E_z = 0$, $H_z = 0$

Q 20. Which of the following is NOT type of waveguide?

- A. Rectangular waveguide
- B. Circular waveguide
- C. Triangular-shape waveguide
- D. Double ridged

Q21. TE mode is correct for which of the following?

- A. Hybrid
- B. Transverse Electromagnetic
- C. Transverse Magnetic
- D. Transverse Electric

Q22. TM mode is correct for which of the following?

- A. Hybrid
- B. Transverse Electromagnetic
- C. Transverse Electric
- D. Transverse Magnetic

Q23. TEM mode is correct for which of the following?

- A. Hybrid
- B. Transverse Electromagnetic
- C. Transverse Magnetic
- D. Transverse Electric

Answer Q 24 – 17 with Circular Waveguide TE modes

χ'_{mn}	$m=0$	$m=1$	$m=2$	$m=3$	$m=4$	$m=5$	$m=6$	...
$n=1$	3.8318	1.8412	3.0542	4.2012	5.3175	6.4155	7.5013	...
$n=2$	7.0156	5.3315	6.7062	8.0153	9.2824	10.5199	11.7349	...
$n=3$	10.1735	8.5363	9.9695	11.3459	12.6819	13.9872	15.2682	...
	:	:	:	:	:	:	:	...

Q24. The table above is called...?

- A. All
- B. Algorithm
- C. Smith Chart
- D. Bessel Function

Q25. In TE_{mn} mode, the subscript 'm' corresponds to which of the following on the table?

- A. None of the below
- B. χ'
- C. Row
- D. Column

Q26. In TE_{mn} mode, the subscript 'n' corresponds to which of the following on the Bessel Function table?

- A. None of the below
- B. χ'
- C. Column
- D. Row

Q27. Which of these values of the zeroes is the dominant mode?

- A. 15.2682
- B. 3.8318
- C. 3.0542
- D. 1.8412

Q28. Which of these following correspond to the dominant mode designation?

- A. TE_{63}
- B. TE_{01}
- C. TE_{21}
- D. TE_{11}

Q29. Which of the following value is the second dominant mode of the table?

- A. 3.8318
- B. 15.2682
- C. 1.8412
- D. 3.0542

Q30. Which of the following correspond to the second dominant mode designation?

- A. TE_{01}
- B. TE_{63}
- C. TE_{11}
- D. TE_{21}

Answer Q 31 – 34 with Circular Waveguide TM modes

χ_{mn}	$m=0$	$m=1$	$m=2$	$m=3$	$m=4$	$m=5$	$m=6$	...
$n=1$	2.4049	3.8318	5.1357	6.3802	7.5884	8.7715	9.9361	...
$n=2$	5.5201	7.1056	8.4173	9.7610	11.0647	12.3386	13.5893	...
$n=3$	8.6537	10.1735	11.6199	13.0152	14.3726	15.7002	17.0038	...
	:	:	:	:	:	:	:	...

Q31. Which of this value of the zeroes is the dominant mode?

- A. 2.4049
- B. 17.0038
- C. 3.8318
- D. 5.1357

Q32. Which of the following correspond to the dominant mode designation?

- A. TM_{01}
- B. TE_{63}
- C. TM_{11}
- D. TM_{21}

Q33. Which of these values is the second dominant mode of the table?

- A. 3.8318
- B. 17.0038
- C. 2.4049
- D. 5.1357

Q34. Which of the following correspond to the second dominant mode designation?

- A. TM_{11}
- B. TE_{63}
- C. TM_{01}
- D. TM_{21}

Q35. W.r.t circular waveguide TM and TEM modes tables above, which of the following is 3rd higher dominant mode?

- A. TE_{21}
- B. TE_{01}
- C. TM_{01}
- D. TE_{11}

Q36. W.r.t Circular Waveguide TM and TEM modes tables above, which of the following is 4th higher dominant mode?

- A. TE₀₁
- B. TE₂₁
- C. TM₀₁
- D. TE₁₁

Answer Q.37 -40 using the information:

An air-filled circular waveguide is to be designed to yield a frequency separation of 1 GHz between the cut-off frequencies of the dominant mode and the next highest mode. Use circular waveguide TM and TEM modes tables of Q. 24 & Q. 31 above.

Q.37. What is the radius of the waveguide?

- A. 2.6896cm
- B. 3.266cm
- C. 4.266cm
- D. 2.6896mm

Q.38. What is the cut-off frequency of the dominant mode?

- A. 3.266GHz
- B. 2.6896GHz
- C. 4.266GHz
- D. 3.266MHz

Q.39. What is the cut-off frequency of the next dominant mode?

- A. 4.266GHz
- B. 2.6896GHz
- C. 3.266GHz
- D. 3.266MHz

Q.40. What is the acronym of the dominant mode and the next?

- A. TE₁₁ and TM₀₁
- B. TE₁₁ and TE₂₁
- C. TE₁₁ and TM₀₁
- D. TM₀₁ and TE₂₁

Q.41. Which of the following is NOT a waveguide component?

- A. Attenuators, Phase shifters
- B. Waveguide bends
- C. Match load
- D. Diode

Answer Q.41- 47 w.r.t to the function of waveguide components:

Q.42. ...can be achieved by using an iris which, can either be on only one side or both sides of the waveguide to balance the system?

- A. Attenuators, Phase shifters
- B. Waveguide bends
- C. Attenuators
- D. Match load

Q.43. ...allow changes in the direction of the transmission?

- A. Attenuators, Phase shifters
- B. Match load
- C. Attenuators
- D. Waveguide bends

Q.44. ...control power levels in a microwave system by partially absorbing the transmitted microwave signal?

- A. Attenuators, Phase shifters
- B. Match load
- C. Waveguide bends
- D. Attenuators

Q.45. ...allow changes in the direction of the transmission?

- A. Attenuators, Phase shifters
- B. Match load
- C. Attenuators
- D. Waveguide bends

Q.46. ...produces a variable shift in phase of the incoming microwave signal.

- A. Attenuators, Phase shifters
- B. Match load
- C. Attenuators
- D. Phase shifters

Q.47. ... combine or divide 3-port signal to 1-port.

- A. Attenuators, Phase shifters
- B. Match load
- C. Attenuators
- D. Magic Tee

Q.48. Which of the following is NOT a Microwave tubes?

- A. Helix TWT
- B. Two cavity Klystron
- C. Magnetron
- D. Duplexers

Q.49. The following is classification of Microwave tubes, except?

- A. Cross-field tubes
- B. linear beam tubes
- C. O-Types or O-Types
- D. Frequencies

Q.50. Klystron is categorized according to what...?

- A. Electrons flow
- B. Transit-time effect
- C. Modulate of electric field
- D. Number of the resonant cavities

Q.51. The following are type Klystron, except...?

- A. Two-cavity klystrons
- B. Multi-cavity klystrons
- C. Reflex klystrons
- D. Electron beam

Q.52. These are Two-Cavities Klystron characteristic, except...?

- A. Efficiency 40%
- B. Power output in CW: up to 500 kW at 10 GHz
- C. Power output in Pulse mode: up to 300 MW at 10 GHz
- D. Power gain: about 0.3 dB

Q.53. Following are components of Two-Cavities Klystron characteristic, except....?

- A. Cathode and Anode
- B. Buncher cavity
- C. Catcher cavity
- D. Semiconductor substrate

Q.54. Which of these is Microwave Semiconductor devices?

- A. Dominant mode
- B. Attenuators
- C. Phase shifters
- D. Gunn Diode

Q.55. Which of the following is a function of a RADAR system?

- A. Gives the range of the target from the Radar station
- B. Gives the radial velocity of target
- C. Detects the presence of target.
- D. All listed options above

Q.56. When two semiconductor materials are brought into contact, the Fermi levels must become aligned to be the same throughout the crystals applies to which of the following.

- A. Transistor
- B. Schottky diode
- C. Phase shifters
- D. PN junction Diode

Q.57. ...have very fast switching times due to their small capacitance.

- A. Transistor device
- B. PN junction Diode
- C. Gunn diode
- D. Schottky diode

Q.58. A slab of N-type GaAs with no PN junctions is...

- A. Transistor device
- B. PN junction Diode
- C. Schottky diode
- D. Gunn diode

Q.59. Which the following best describe what a fibre optics is...?

- A. It is made of low-loss materials such as silica or glass.
- B. All of the listed A, C & D options.
- C. It has a central core in which the light is guided.
- D. Cylindrical dielectric waveguide.

Q.60. Fibre optics core and cladding have different refractive indices... which of the following is TRUE about the core and cladding of a fibre optics...?

- A. Cladding also have a refractive index, n_2 .
- B. All of the listed options below
- C. refractive index n_1 slightly higher than n_2 .
- D. Core has different refractive index, as n_1

Q.61. Which of the following can be used to discuss the geometry of optical fibre?

- A. Maxwell's law.
- B. Snell's law.
- C. Faraday's law.
- D. Biot Savart law.

Q.62. In RADAR system, the term pulse repetition time (abbreviated PRT) is mathematically giving as time:

- A. For pulse to reach the receiving antenna
- B. For pulse transmission to hit the target and return
- C. The beginning of first pulse to the beginning of the next
- D. All the above

Q.63. Radar is an Acronym for RAdio Detection And Ranging

- A. Radio Development And Range.
- B. RAdio Detection And Ranging.
- C. Radiology Device.
- D. None of the above.

Q.64. The two basic radar systems are...

- A. Two-cavity and Multi-cavity.
- B. Monostatic and Bistatic.
- C. O-Types and M-Types.
- D. None of the above.

Q.65. The following is a function of RADAR system, except

- A. Detects the presence of target
- B. None of the listed value
- C. Gives the radial velocity of target.
- D. Gives the azimuth and elevation angle of the target

Q.66. A full duplex antenna RADAR system is...

- A. Automatic Radar
- B. Single antenna for both transmitting and receiving
- C. Manual radar system.
- D. None of the above

Q.63. RADAR system can be use in the Air Traffic Control to...?

- A. Demobilised the moving target approaching
- B. To safely control air traffic in the airports/enrooted
- C. Broadcast to the passenger's flight's information
- D. All the above listed options

Q.64. RADAR range can be measured in which unit?

- A. Nautica Miles (nmi)
- B. Kilometres (Km)
- C. All the listed options
- D. Metres (m)

Q.65. In RADAR terms, PRF stands for...?

- A. Periodic Rare Frequency
- B. Phase Rate Frequency)
- C. Pulse Repetition Frequency
- D. None of the above

Q.66. In RADAR terms, PRT stands for...?

- A. Periodic Rare Time
- B. Phase Rate Time
- C. Pulse Repetition Time
- D. None of the above

Q.67. Which of the following body are tasked with the allocation and regulating the frequency bands for Radar system?

- A. Radio Broadcasting
- B. Military Defence Services
- C. International Telecommunication Union (ITU)
- D. Airport Authority

Q.68. The following are application of a RADAR systems, EXCEPT?

- A. Air Traffic Control
- B. Military Defence
- C. Radio Broadcasting Services
- D. Remote Sensing

Q.69. Law Enforcement & Highway Safety use RADAR system to?

- A. For warning of pending collision
- B. Actuating air bag or warning of obstruction
- C. To measure the speed for enforcing speed limits
- D. Demobilised the moving target

Q.70. Optical fibre structure, $n_1 > n_2$ then it is always that

- A. $f_1 \pm f_2$.
- B. $f_1 > f_2$
- C. $f_1 < f_2$.
- D. $f_1 = f_2$

Q.71. Satellites are specifically made for purpose?

- A. Detecting.
- B. Alignment.
- C. Telecommunication
- D. All the above.

Q.72. They are responsible for providing services

- A. On ocean and sea.
- B. Alignment on space
- C. an assigned region on the earth
- D. All the above.

Q73. Power and bandwidth of satellites system depend upon...

- A. Size of the footprint
- B. Complexity of the traffic control
- C. protocol schemes and Cost of ground stations
- D. All the above.

Q. 74. Satellite's antenna patterns must be designed to...

- A. To accommodate traffic protocols
- B. To address the drifting problem on space
- C. **To best cover the designated geographical area**
- D. None of the above.

Q. 75. Which the following is not an orbits satellites system?

- A. Circular
- B. Elliptical
- C. Rectangular
- D. None of the above.

Q. 76. The force trying to pull satellite away from the earth is...?

- A. Centripetal
- B. gravitational
- C. Centrifugal
- D. None of the above.

Q. 77. The force trying to pull satellite towards the earth is...?

- A. Centrifugal
- B. Atmospheric
- C. Centripetal
- D. All the above.

Q. 78. Which of the following is NOT responsible for satellite drifting away from its designated orbit?

- A. Centripetal
- B. gravitational
- C. Centrifugal Force
- D. Weight of the satellite.

Q. 79. For satellite to remain on its designated orbit, the...

- A. Centripetal = 0
- B. gravitational > Centripetal or Centrifugal
- C. Centripetal Force = Centrifugal Force
- D. satellite weight on space must be very small.

Q. 80. Distance of a satellite to the earth's surface depends on...

- A. Its capacity to communicates.
- B. Its footprint size on earth
- C. its rotation frequency
- D. None of the above.

Q. 81. Inclination angle of 0-degrees means that the satellite is...

- A. Satellite not yet launch on space
- B. Same azimuthal and elevation angle
- C. Exactly above the equator
- D. None of the above.

Q. 83. Satellite closest point to the earth is called the...

- A. Apogee
- B. Satellite orbit
- C. Perigee
- D. All the above.

Q. 84. Satellite furthest point to the earth is called the...

- A. **Perigee**
- B. Satellite orbit
- C. Apogee
- D. None of the above.

Q. 85. All these parameters are important in satellite communication except...

- A. **inclination angles.**
- B. **elevation angles.**
- C. Wight of the satellite
- D. None of the above.

Q. 86 Which of the following is NOT a satellite orbit?

- A. LEO.
- B. MEO.
- C. NEO
- D. GEO.

Q. 87 GEO satellite are called geostationary satellite because?

- A. Are NOT synchronous with respect to earth.
- B. Are Bigger satellites with heavy payloads
- C. Are synchronous with respect to earth
- D. All the above

Q88. Which of the following defined the distance of GEO satellites from the earth surface?

- A. 37,86 km.
- B. 10,000 km.
- C. **500-1500 km.**
- D. None of the above.

Q89. Which of the following defined the distance of MEO satellites from the earth surface?

- A. 37,86 km.
- B. 10,000 km.
- C. **500-1500 km.**
- D. None of the above.

Q90. Which of the following defined the distance of LEO satellites from the earth surface?

- A. 37,86 km.
- B. 10,000 km.
- C. **500-1500 km.**
- D. None of the above.

Q91. ITU requires international coordination and planning for frequency allocation to...?

- A. Satellite launching
- B. Satellite decommission.
- C. Satellite frequency.
- D. Satellite control.

Q. 92. All are conditions leading to GEO, EXCEPT...?

- A. **Should be placed approximated to 36,000 km**
- B. **Must travel in the rotational speed of earth.**
- C. **Inclination of satellite w.r.t earth must be 0°**
- D. None of the listed conditions.

Q93. In frequencies allocation for satellite services, the whole world is divided into how many regions?

- A. One region
- B. Two regions.
- C. Three regions.
- D. Four regions.

Q94. GEO satellites constellation required only ... to cover whole earth surface?

- A. 27 satellites.
- B. 9 satellites.
- C. 3 satellites.
- D. None of the above.

Q95. Which of the following is NOT the applications of satellite system?

- A. Global Mobile Communications
- B. Radio and TV broadcast.
- C. Highway/Law Enforcement
- D. Navigation services.

Q96. For LEO satellites, transmission rates of... ..can be enough for voice communication?

- A. about 2400 bit/s.
- B. about 240 bit/s.
- C. about 24 bit/s.
- D. None of the above.

Q97. Smaller footprints of ... satellites allow for better frequency reuse, like the cellular networks?

- A. GEO
- B. MEO
- C. LEO.
- D. All the above.

Q98. Which of the following is NOT the application of satellites?

- A. Global Telephony system
- B. Connecting remote areas
- C. Radio and TV broadcast
- D. None of the above.

Q99. In frequencies allocation for satellite services by ITU, Europe, Africa, and Mongolia is which of the following region?

- A. Region 1
- B. Regions 2
- C. Regions 3
- D. All of the above

Q100. In frequencies allocation for satellite services by ITU, North and South America and Greenland is which of the following?

- A. Region 1
- B. Regions 2
- C. Regions 3
- D. All the above

Q100. In frequencies allocation for satellite services by ITU, North and South America and Greenland is which of the following region?

- A. Region 1
- B. Regions 2
- C. Regions 3
- D. All the above

Q101. In frequencies allocation for satellite services by ITU, Asia (excluding region 1 areas), Australia and south-west Pacific is which of the following region?

- A. Region 1
- B. Regions 2
- C. Regions 3
- D. All the above

Q102. Nigeria falls in which of the following region?

- A. Region 1
- B. Regions 2
- C. Regions 3
- D. All the above

Q103. Which of the following frequency band is NOT allocated for satellite services?

- A. L-band 1.0 – 2.0GHz.
- B. C-band 4.0 – 8.0GHz.
- C. mm-Wave Frequencies
- D. Ku-band 12 – 18GHz.

104. Which of the following is NOT satellite services?

- A. Fixed satellite service
- B. Broadcast satellite service
- C. Mobile satellite service
- D. None of the above

105. Which of the following satellite services provides Links for existing Telephone Networks?

- A. Fixed satellite service
- B. Broadcast satellite service
- C. Mobile satellite service
- D. None of the above

106. Which of the following satellite services provides direct broadcast to homes?

- A. Fixed satellite service
- B. Broadcast satellite service
- C. Mobile satellite service
- D. None of the above

107. Which of the following satellite services provides land mobile?

- A. Fixed satellite service
- B. Broadcast satellite service
- C. Mobile satellite service
- D. None of the above

SECTION B: ESSAY

Instruction: Answer all QUESTION from this section.

TOPIC I Waveguide:

Q.1 What is electromagnetic wave?

Q.2 Show that, for an EM wave $E_y(x, y, z, t) = E_y(z, t)$ i.e., with the propagating in the z direction, and with no dependence in the x or y direction is mathematically equal to

$$\frac{\partial^2 E_y(z)}{\partial z^2} + k^2 E_y(z) = 0$$

Q.3 List any four (4) properties of electromagnetic wave and define any two (2).

Q.4 Mathematically, demonstrate the relationships among the listed properties in Q1.b above.

Q.5 What is waveguides, list and draw the 5 types of waveguides.

Q.6 Assuming, an ideal parallel-plate waveguide which is infinite in extent in the y and z directions (with perfectly conducting plates i.e., ϵ and μ are simple constants, lossless. i.e., $\alpha = 0$, μ and ϵ are real). Using Maxwell's equation, show that:

$$\begin{aligned} \bar{\gamma} E_y &= -j\omega\mu H_x & \bar{\gamma} H_y &= j\omega\epsilon E_x \\ -\bar{\gamma} E_x - \frac{\partial E_z}{\partial x} &= -j\omega\mu H_y & -\bar{\gamma} H_x - \frac{\partial H_z}{\partial x} &= j\omega\epsilon E_y \\ \frac{\partial E_y}{\partial x} &= -j\omega\mu H_z & \frac{\partial H_y}{\partial x} &= j\omega\epsilon E_z \end{aligned}$$

Q.7 What are the three categories' solutions of guided-wave equations?

Q.8 Mathematically derive the expression for field solutions of TE, TM, and TEM waves modes

Q.9 Using the values of zeros as presented in the Bessel function and Bessel function derivative tables below, copy and complete the table

TE modes

χ'_{mn}	$m=0$	$m=1$	$m=2$	$m=3$	$m=4$	$m=5$	$m=6$	...
$n=1$	3.8318	1.8412	3.0542	4.2012	5.3175	6.4155	7.5013	...
$n=2$	7.0156	5.3315	6.7062	8.0153	9.2824	10.5199	11.7349	...
$n=3$	10.1735	8.5363	9.9695	11.3459	12.6819	13.9872	15.2682	...
	:	:	:	:	:	:	:	...

TM modes

χ_{mn}	$m=0$	$m=1$	$m=2$	$m=3$	$m=4$	$m=5$	$m=6$	...
$n=1$	2.4049	3.8318	5.1357	6.3802	7.5884	8.7715	9.9361	...
$n=2$	5.5201	7.1056	8.4173	9.7610	11.0647	12.3386	13.5893	...
$n=3$	8.6537	10.1735	11.6199	13.0152	14.3726	15.7002	17.0038	...
	:	:	:	:	:	:	:	\

Note: From the tables above, the 1st dominant mode is the TE₁₁ mode, followed in order by the TM₀₁ mode, then the TE₂₁ mode, and the TE₀₁ mode. Hence the complete table is given below as:

S/No.	Order	Mode	Designation	Values of Zeroes
1.	1 st Dominant	TE	TE ₁₁	1.8412
2.	2 nd Dominant			
3.	3 rd Dominant			
4.	4 th Dominant			

Q.10 Design an air-filled circular waveguide yielding a frequency separation of 1 GHz between the cut-off frequencies of the dominant mode and the next highest mode.

Q.11 List, draw and briefly explain any ten (10) components of waveguides.

Q.12 Define a Microwave tube, list and briefly explain its classification.

Q.13 List and briefly explain the two types of Microwave tube.

Q.14. Write the characteristics of the two listed Microwave tubes in Q.13 above.

Q.15 Draw a complete diagram of any one of the klystrons listed in Q.13, furthermore, list and briefly explain any five of its components.

Q.16. List and explain any six (6) Microwave tube semiconductors.

Q.17. Define a fibre optics, and briefly discussed the propagation principle using Snell's law

TOPIC I Waveguide:

Q.1 With aid of a simple diagram, define is a RADAR system.

Q. 2. List and briefly explain any five (5) RADAR system terms/acronyms.

Q.3 List any five (5) function of a RADAR system

Q.4 With the aid of block diagram, explain a full duplex RADAR system and all the respective blocks.

Q.5 List nine (9) FIELD of applications of a RADAR system and states any 3 sub-applications of these fields.

Q.6. Derive mathematically, the simple form of RADAR equation (range performance) and define all the respective parameters.

Q7. Calculate the maximum range of a radar which operates at a frequency of 20GHz, peak pulse power of 1.2Mwatts. If the antenna effective area is 10m² and the area of the target is 40m² and the minimum receivable power of 0.2n watts.

Topic III: Satellite Communications

Q.1 What is a satellite system?

Q.2. list the three (3) properties in which Power and bandwidth of satellites depend upon

Q.3. List and briefly explain the different satellites orbit around the earth.

Q.5. Show mathematically, for the distance (r) of the satellite to the centre of the earth is:

$$r = [g \times R^2 / (2\pi f)^2]^{1/3}$$

Q.6. In not more than one paragraph, state what is the dependence of distance of a satellite to the earth's surface and why?

Q.7. Explain in detail, the following satellite orbits.

i. GEO

ii. MEO

iii. LEO

Q.8. List the three (3) conditions which lead to geostationary satellites.

Q.9. What are factors which make satellites (especially GEO) shift from the ideal geostationary orbit.

Q.10. List and briefly explain, the advantages and disadvantages the following satellites system.

i. GEO satellites

ii. MEO satellites

iii. LEO satellites

Q.11 List any eight (8) applications of a satellite system and briefly explain any four (4).

Q.12 Why is it that, frequencies allocation for satellite services is a complicated process?

Q.13. With respect to the satellites system deployment and operations, what are the conditions of implementing frequency planning?

The condition of implementing frequency planning of satellites system deployment and operations is that the world is divided into three regions, this requires international coordination and planning. These regions include:

- ✓ Region1: Europe, Africa, and Mongolia
- ✓ Region 2: North and South America and Greenland
- ✓ Region 3: Asia (excluding region 1 areas), Australia and south-west Pacific.

Q.14 List any ten (10) frequency band in frequency spectrum.

Frequency Band (GHZ) Designations:

- VHF: 01-0.3
- UHF: 0.3-1.0
- L-band: 1.0-2.0
- S-band: 2.0-4.0
- C-band: 4.0-8.0
- X-band: 8.0-12.0
- Ka-band: 18.0-27.0 (Ka is Above K Band)
- V-band: 40.0-75.0
- W-band: 75-110
- mm-band: 110-300
- μ m-band: 300-3000

Q.15. List five (5) frequency band and their respective designation allocated for satellites systems.

The frequencies allocated to these satellites: Frequency Band (GHZ) Designations:

<u>Frequency Band (GHZ)</u>	<u>Designations:</u>
VHF: 01-0.3	Mobile & Navigational Satellite Services
L-band: 1.0-2.0	Mobile & Navigational Satellite Services
C-band: 4.0-8.0	Fixed Satellite Service
Ku-band: 12.0-18.0	Direct Broadcast Satellite Services

Q.16. One of the frequency bands allocated for satellites system is the HF band, list and briefly explain the band satellite services.

Some of the applications of the bands in satellites services are listed below.

- Fixed satellite service: Provides Links for existing Telephone Networks Used for transmitting television signals to cable companies.
- Broadcasting satellite service: Provides Direct Broadcast to homes e.g., Live broadcast of political parties rallies and conventions etc
- Mobile satellite services: This includes services for: Land Mobile
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